

LeafFit: Plant-Asset Creation from 3D Gaussian Splatting

Chang Luo Umetani Nobuyuki

The University of Tokyo



Converting 3DGS-captured Plant to Asset-ready Mesh



Input: 3DGS



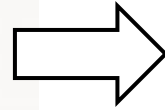
Output: Mesh

Implicit method's output are heavy



3DGS Captured Plant

Implicit
Methods

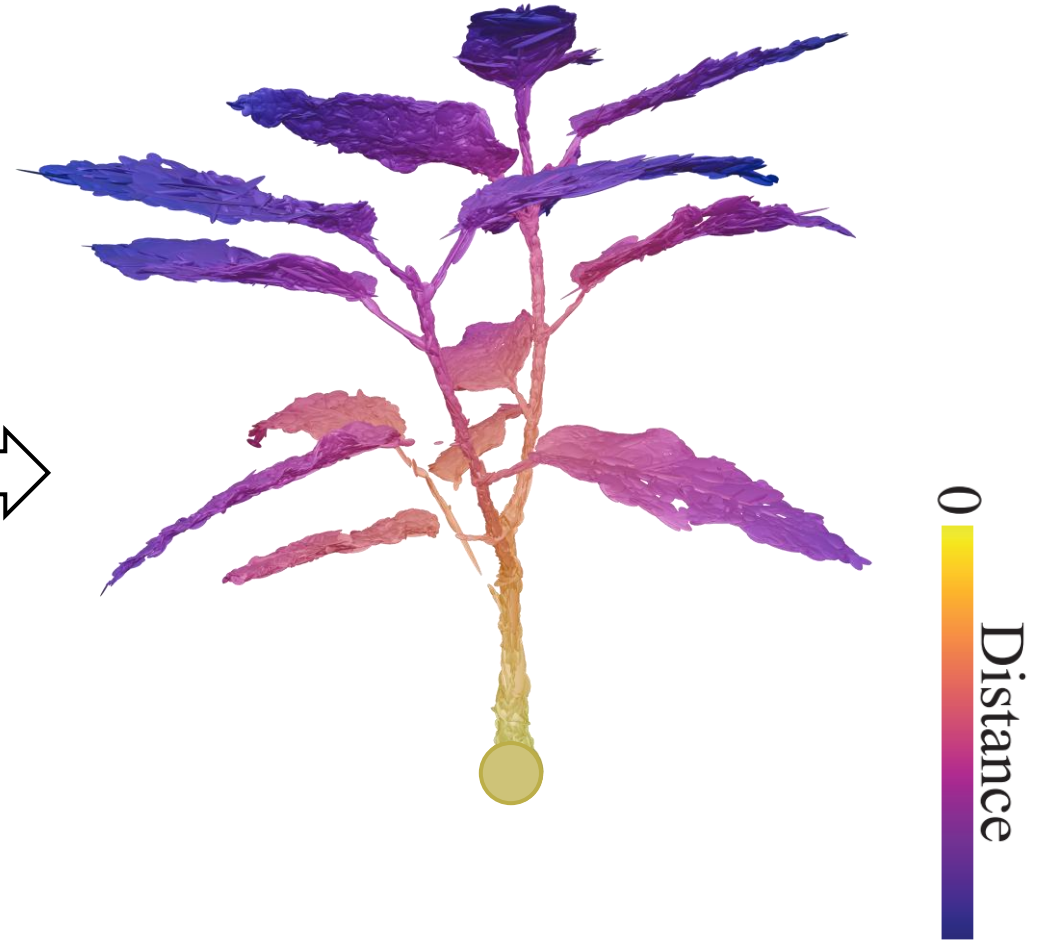
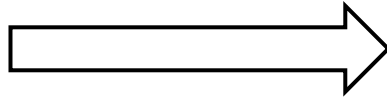


#Verts	1,856,309
#Faces	3,737,746

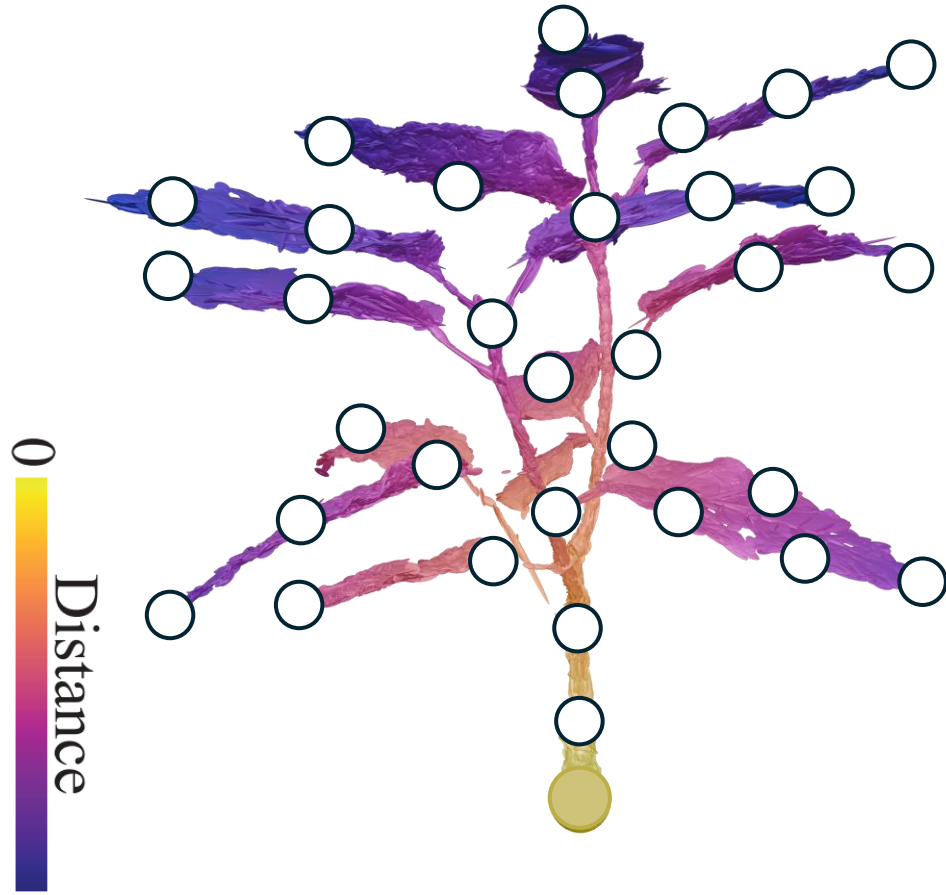
≈ **76 MB**

Output Mesh

Estimate Geodesic Distance Fields

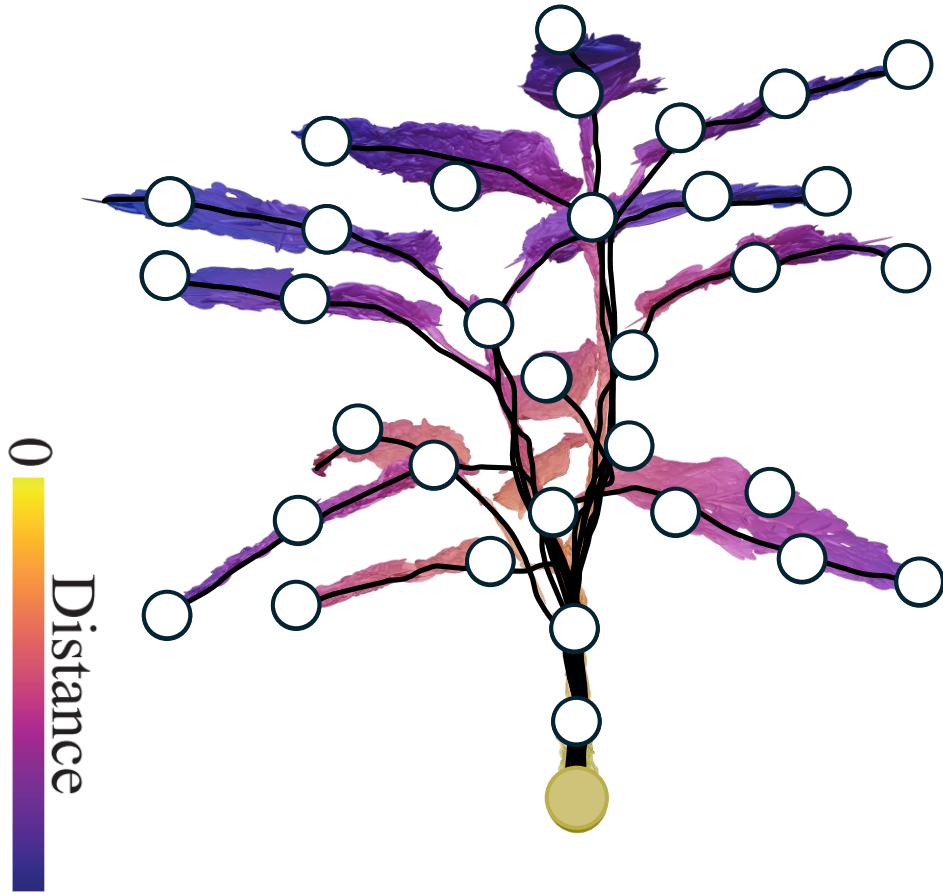


Down-sample



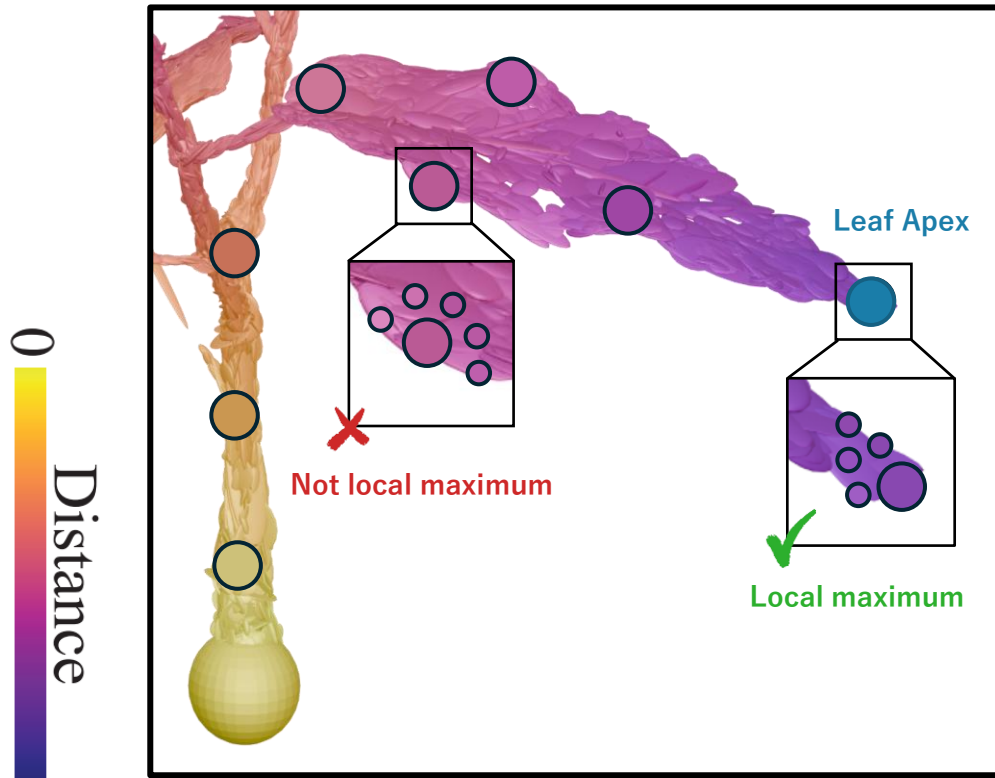
- Get sparse points ○ by down-sampling

Measure distance to root



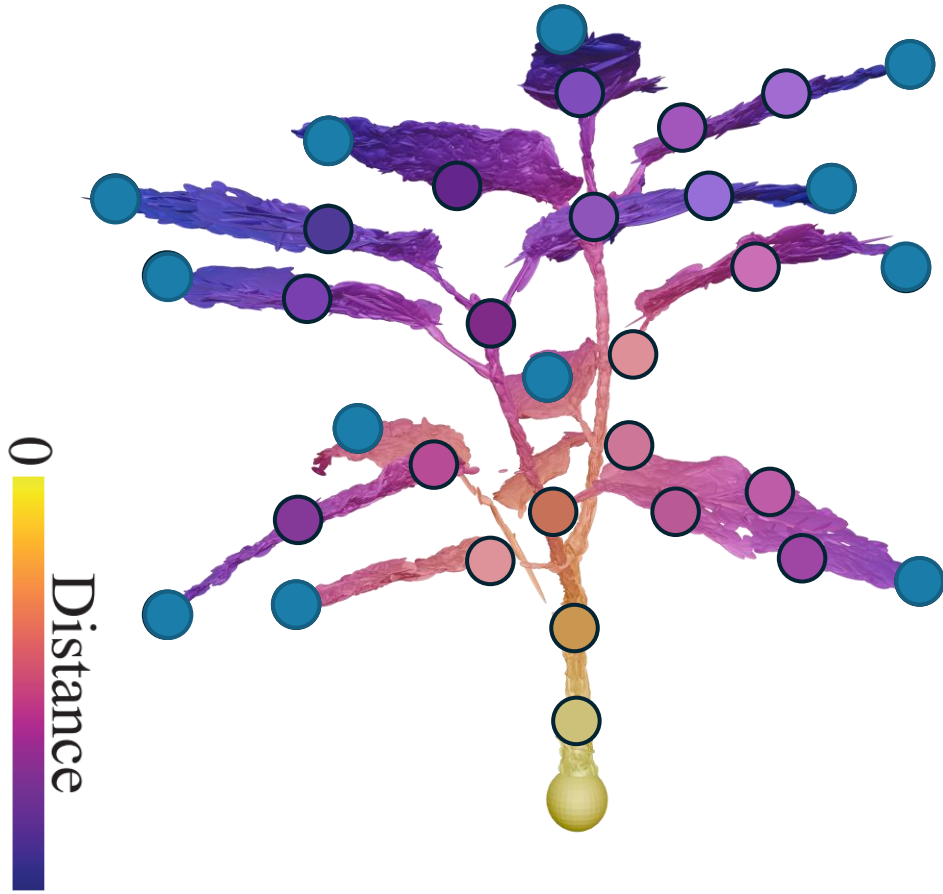
- Get sparse points $\bigcirc$ by down-sampling
- Get the geodesic distance to **root** $\bullet$

Determine leaf apexes



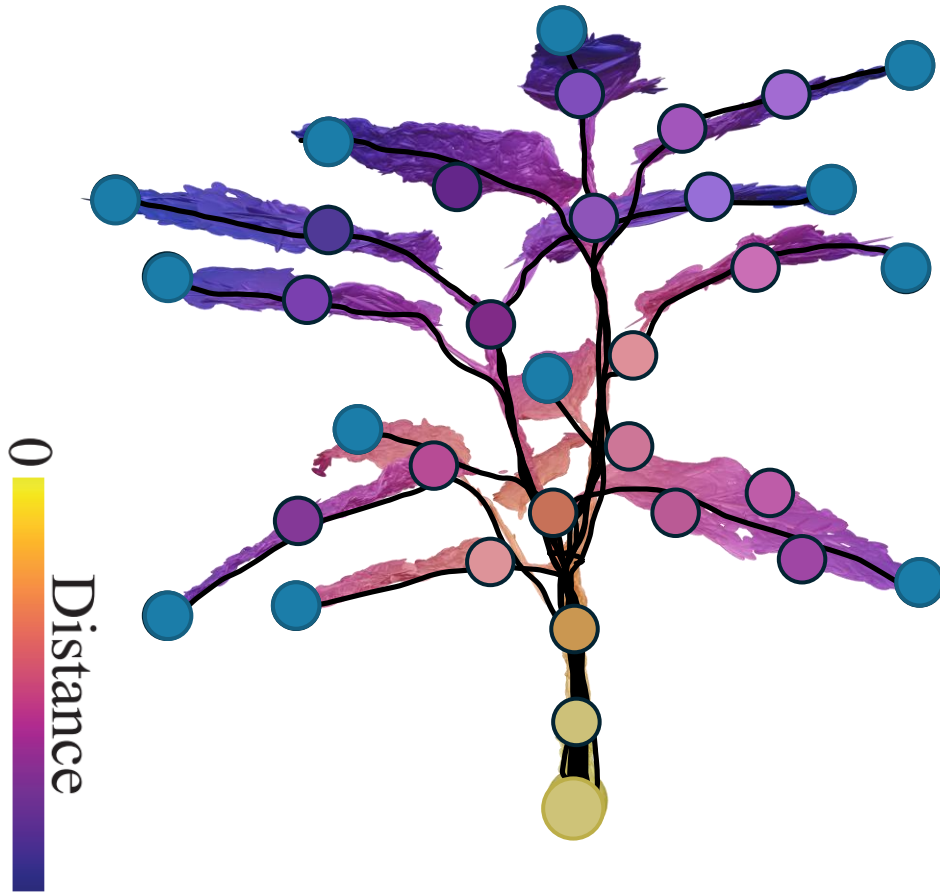
- Get sparse points ○ by down-sampling
- Get the geodesic distance to **root** ●
- Check neighbors to find leaf apices ●

Determine leaf apexes

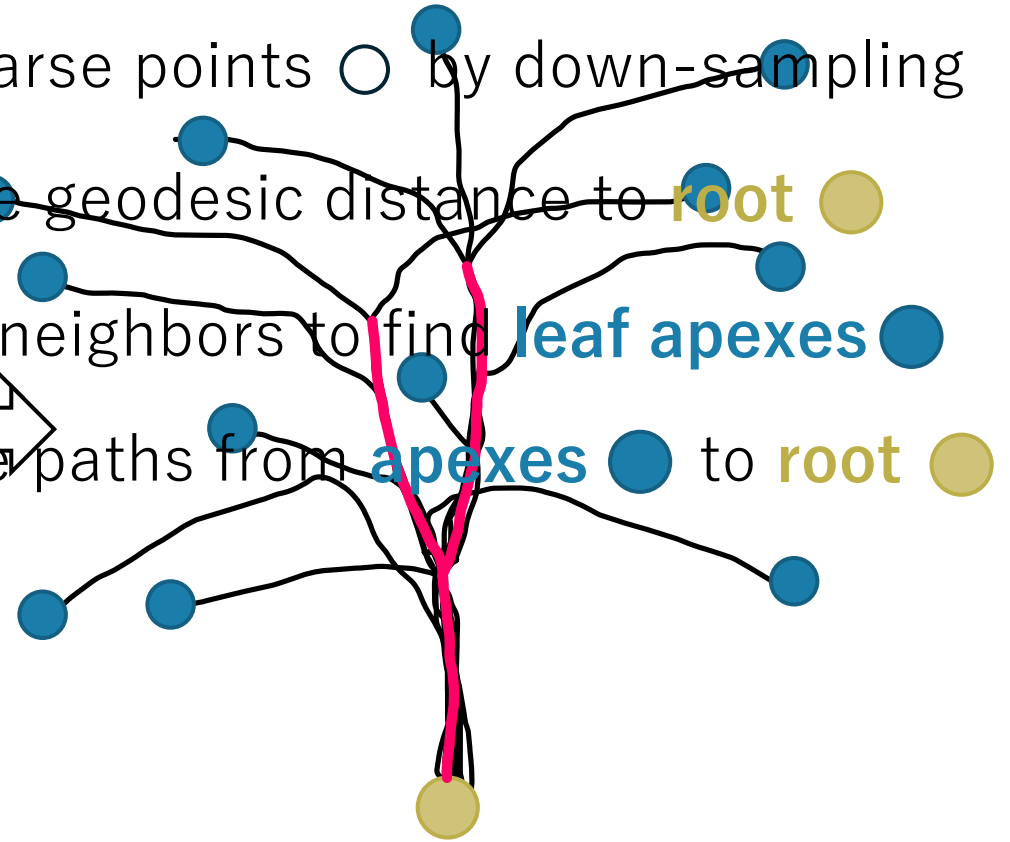


- Get sparse points ○ by down-sampling
- Get the geodesic distance to **root** ●
- Check neighbors to find **leaf apexes** ●

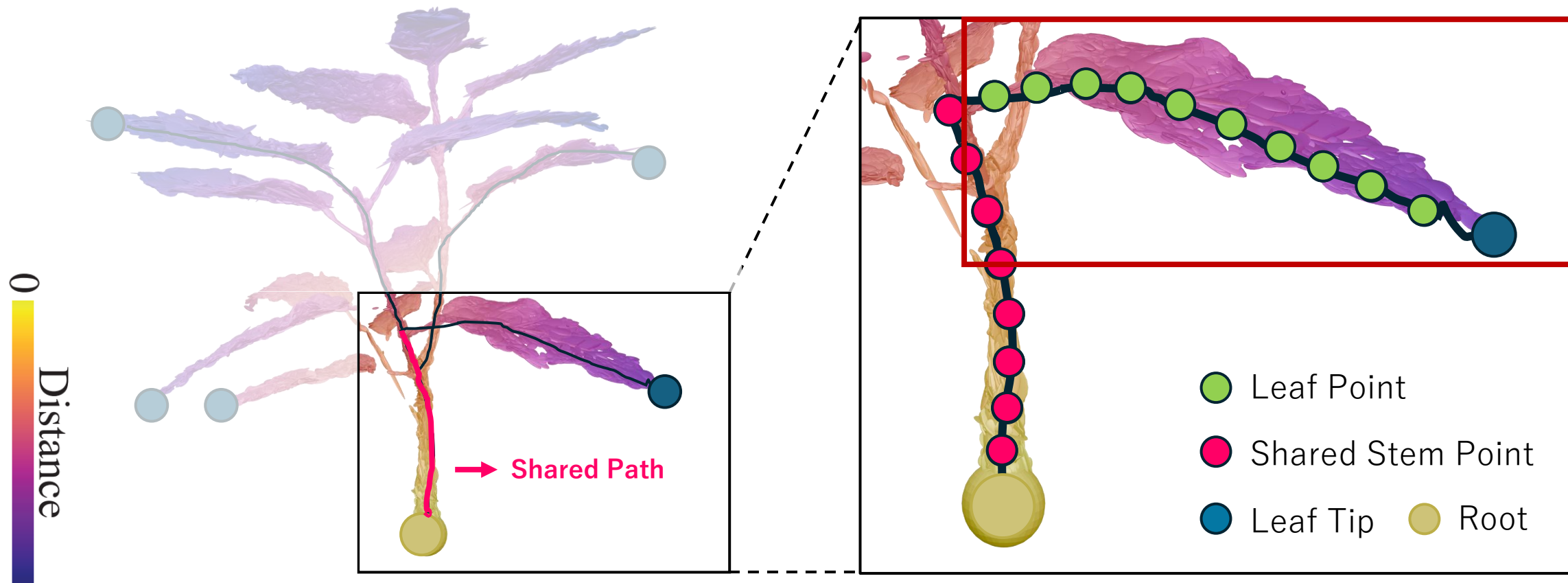
Get Apex-Root pathes



- Get sparse points $\bigcirc$ by down-sampling
- Get the geodesic distance to **root** $\bullet$
- Check neighbors to find **leaf apexes** $\bullet$
- Get the paths from **apexes** $\bullet$ to **root** $\bullet$

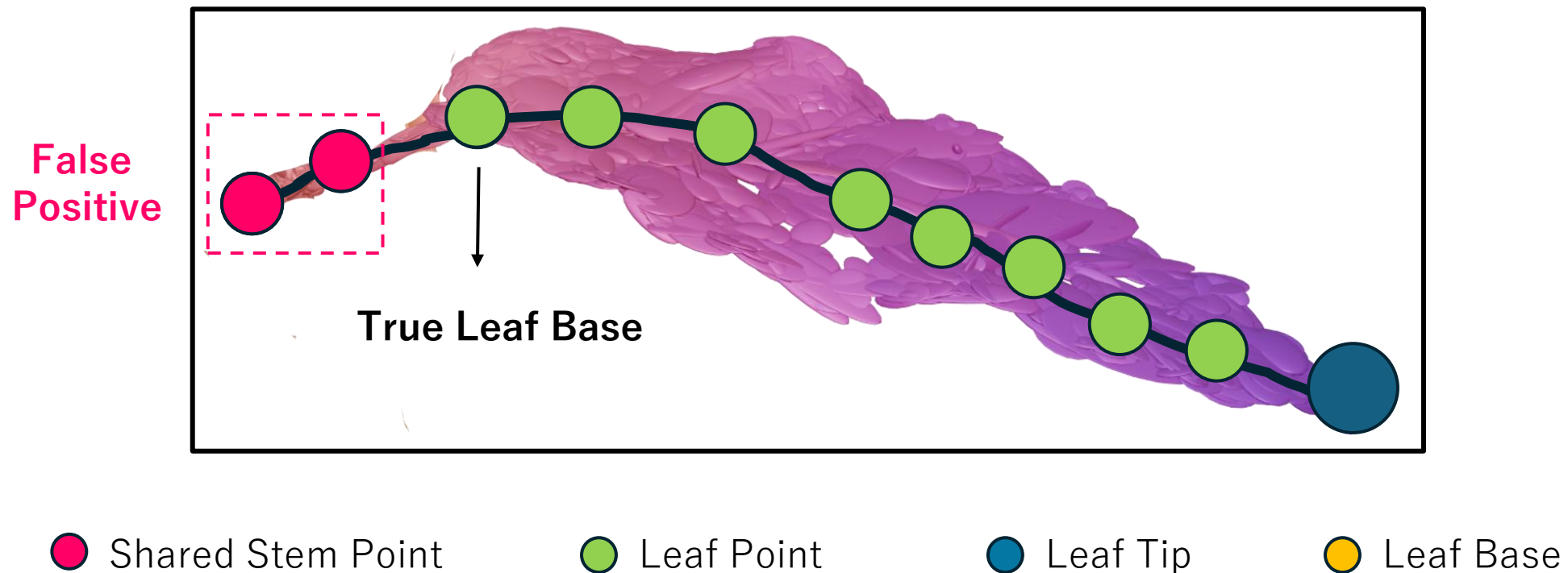


Leaf-stem Segmentation



Leaf-stem Segmentation

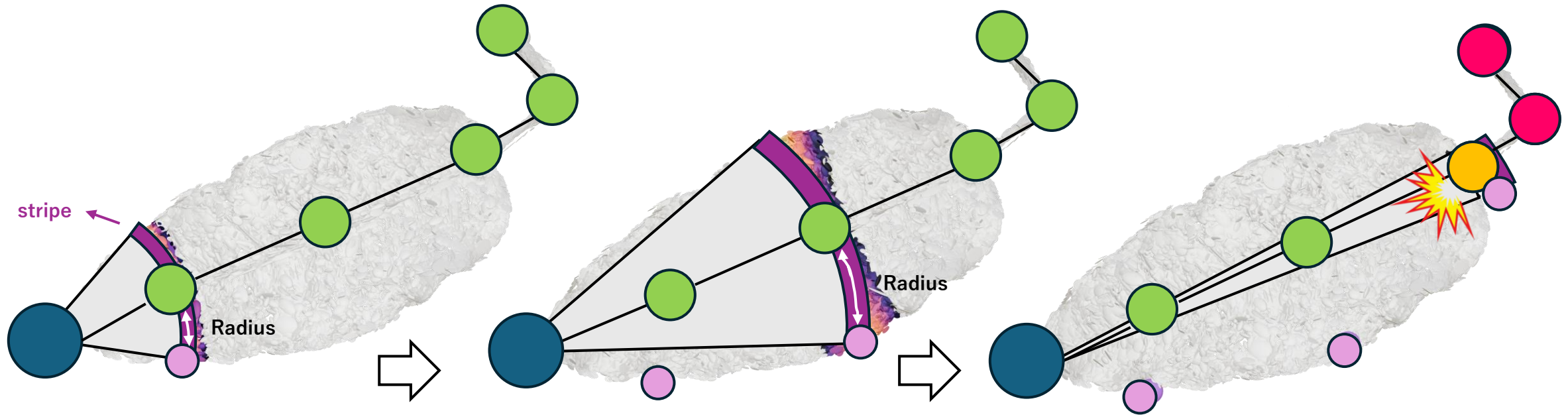
Some points are misclassified as **leaf points**.



Find real leaf base

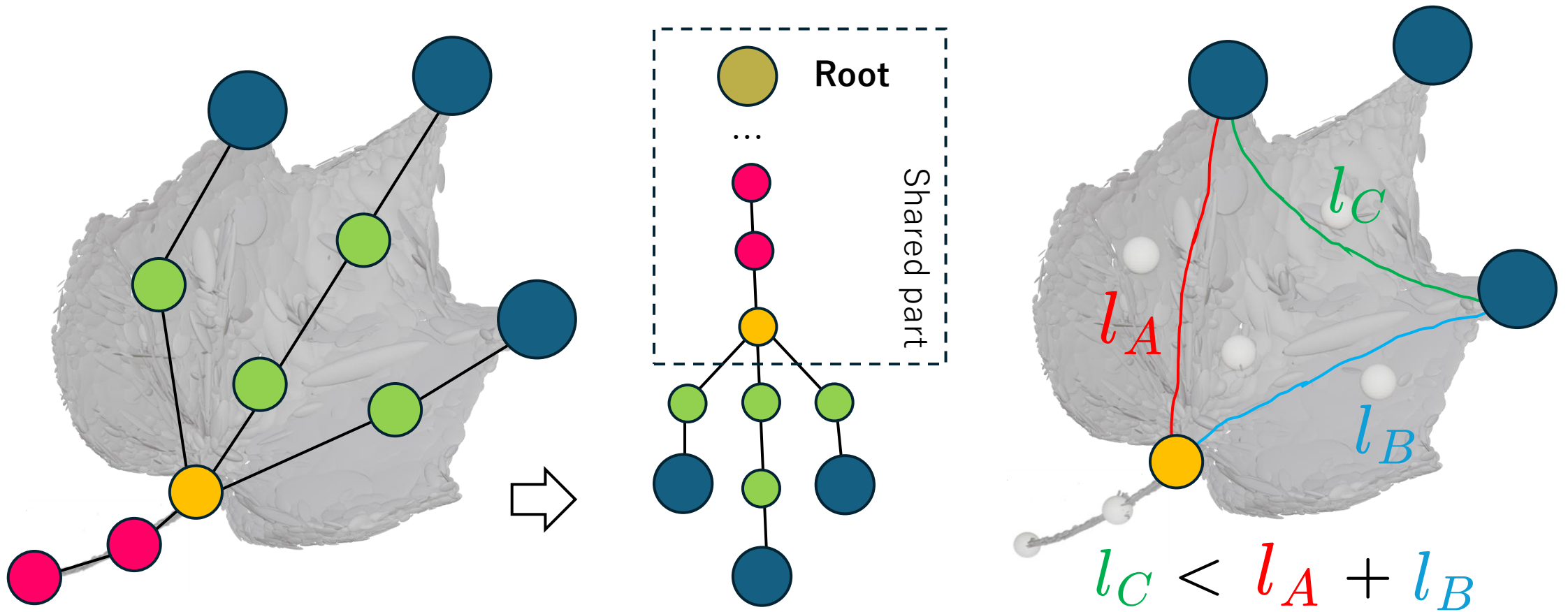
Radius is computed farthest distance from center (leaf points).

Leaf base has the smallest distance.



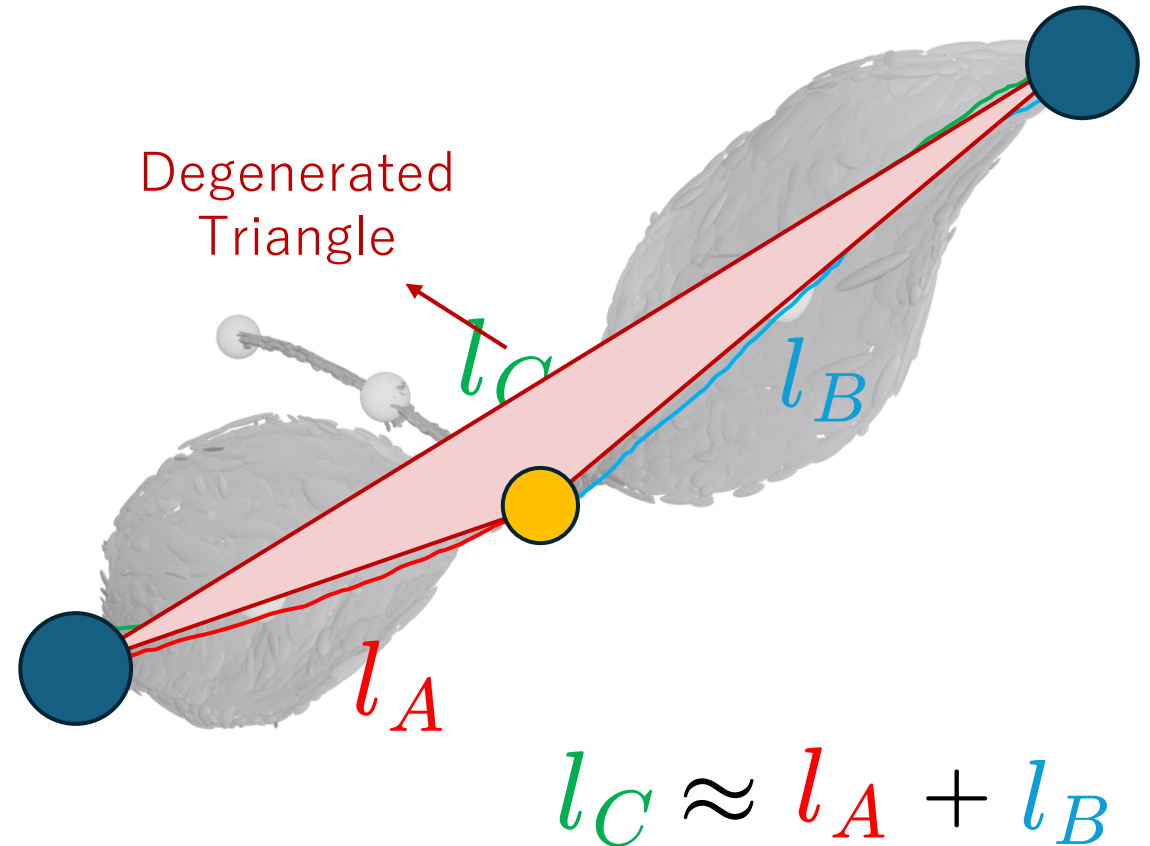
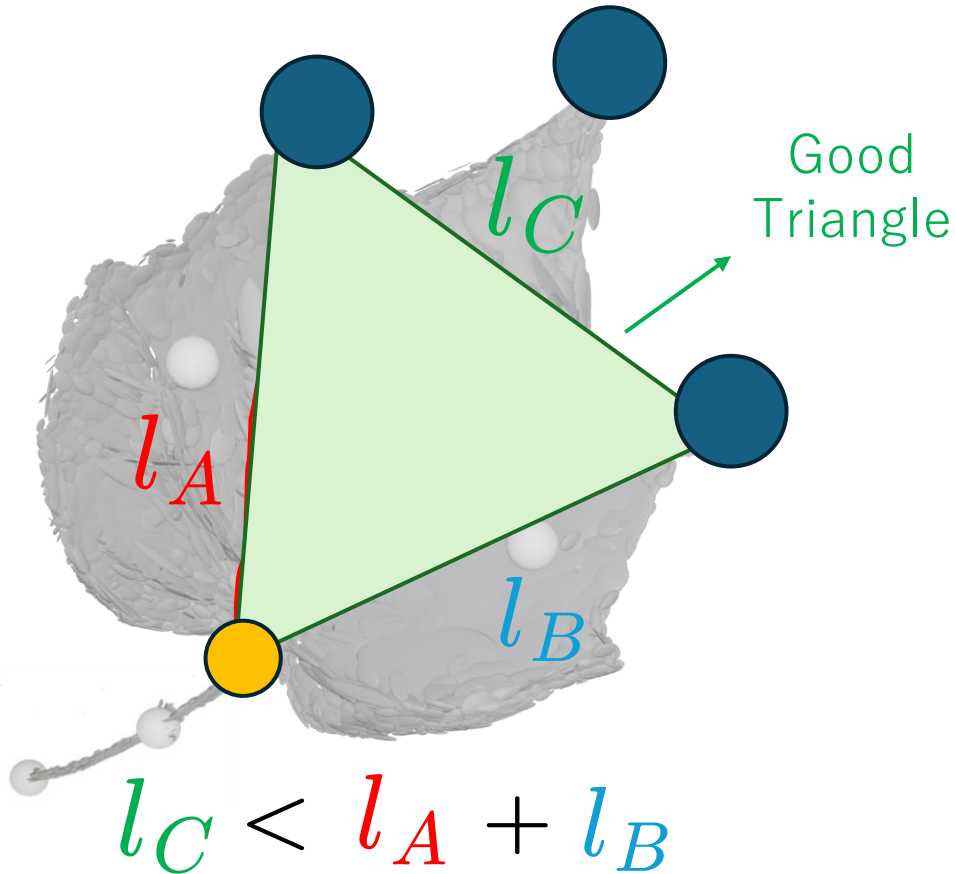
Tip Grouping for complicated leaves

- We group **tips** to where triangle inequality is unsatisfied.



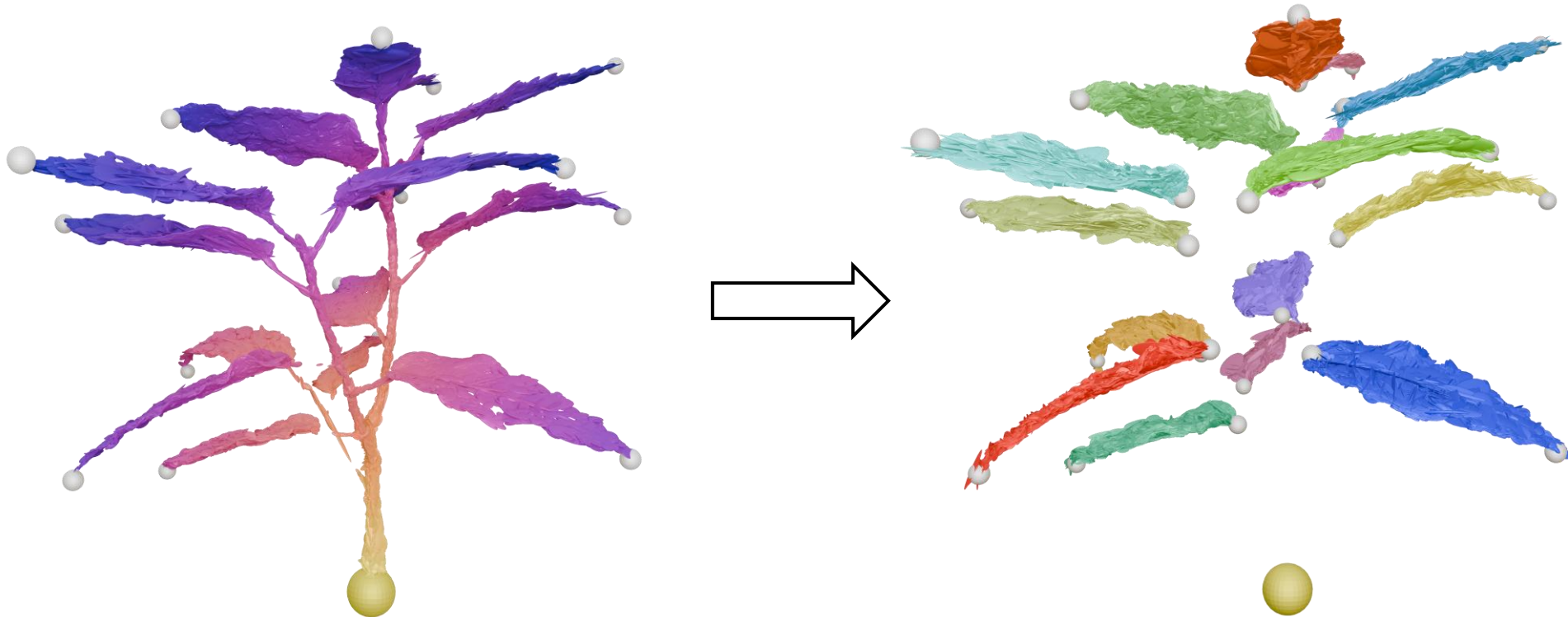
Tip Grouping for complicated leaves

Inequality of Triangle is used to group leaf tips.



Automatic Segmentation

Leaves are well segmented now.

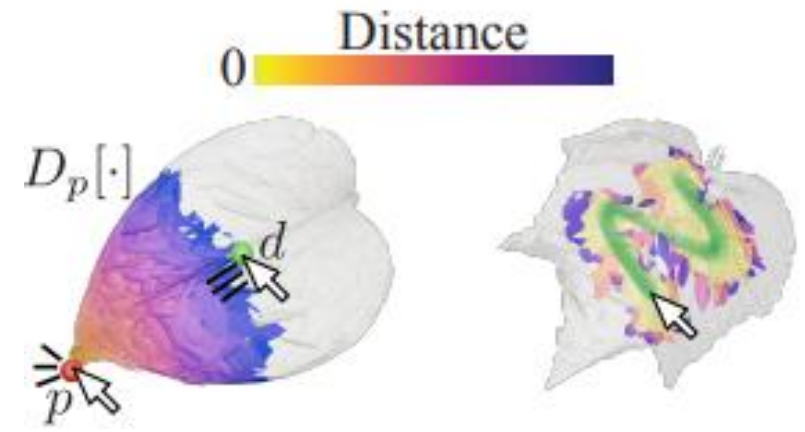
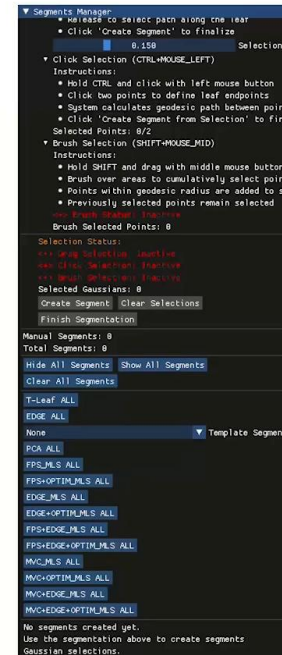


Manual Segmentation

Manual fallback is also possible.

Drag Tool

x8

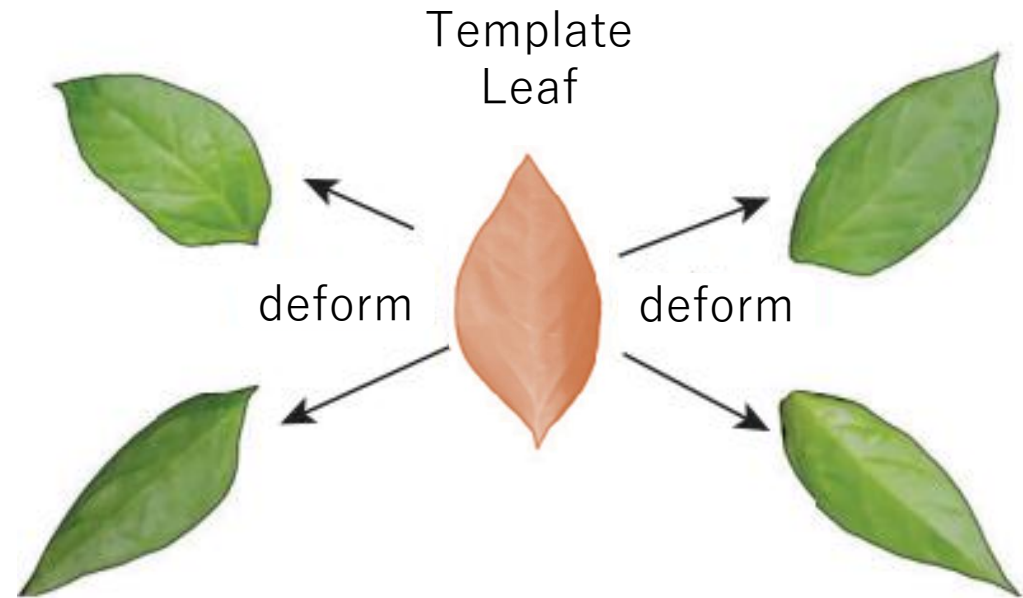


Drag

Brush

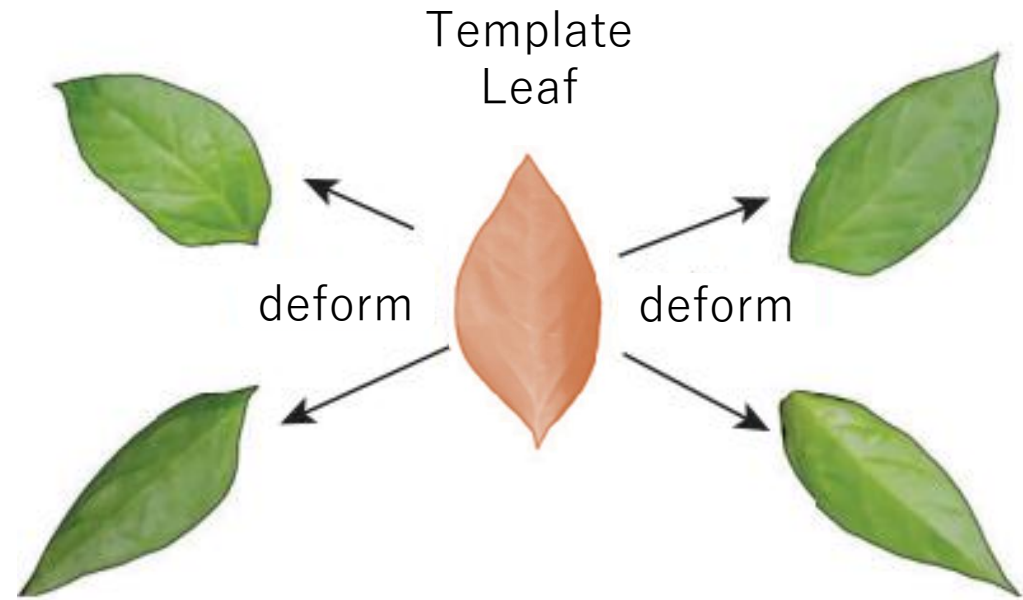
Compression using Similarity

Leaves are similar, take this inherent similarity to compress.



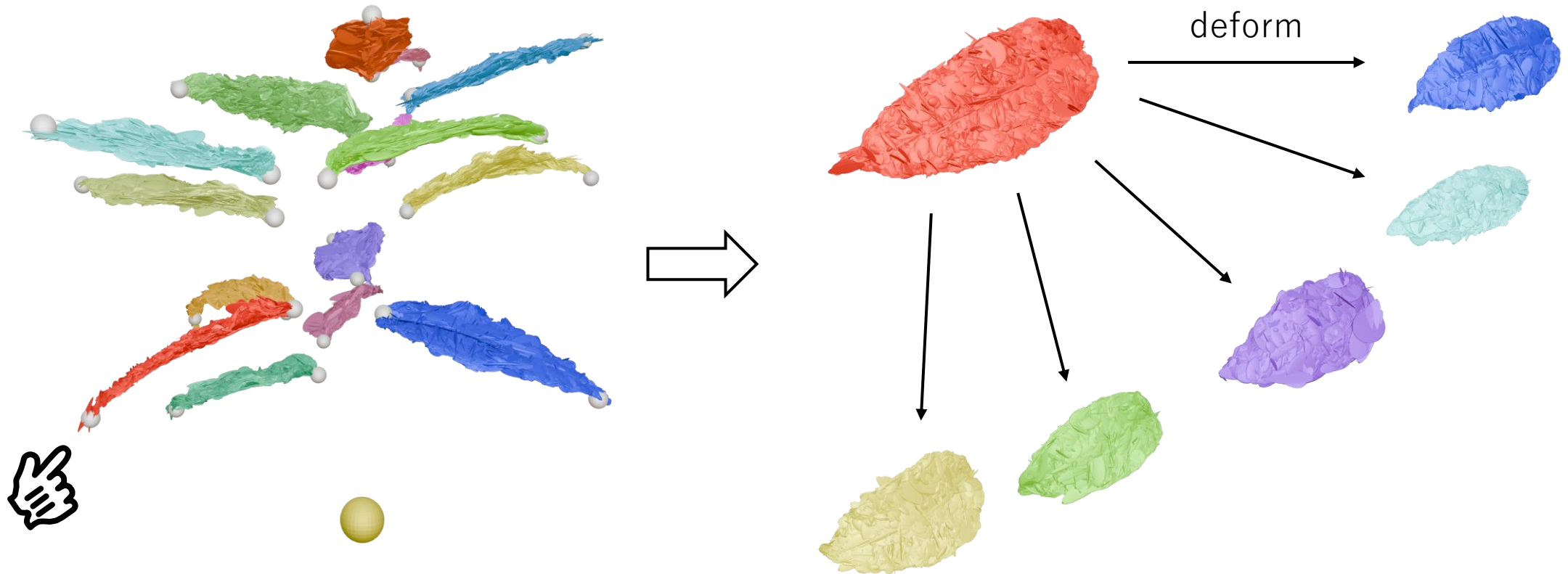
Compression using Similarity

Leaves are similar, take this inherent similarity to compress.



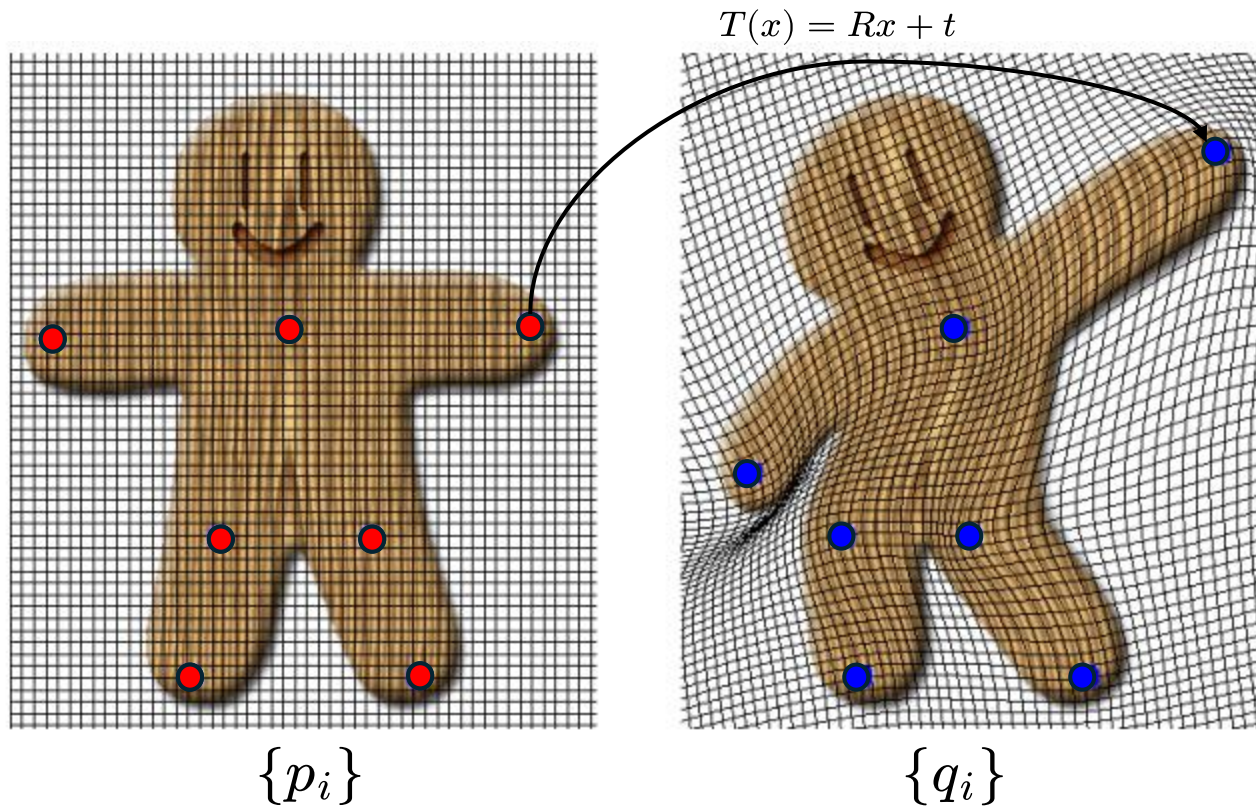
Similarity

Leaves are similar, take this inherent similarity to compress.



Moving Least Squares

Moving Least Squares [Schaefer+06] optimizes local transformation.



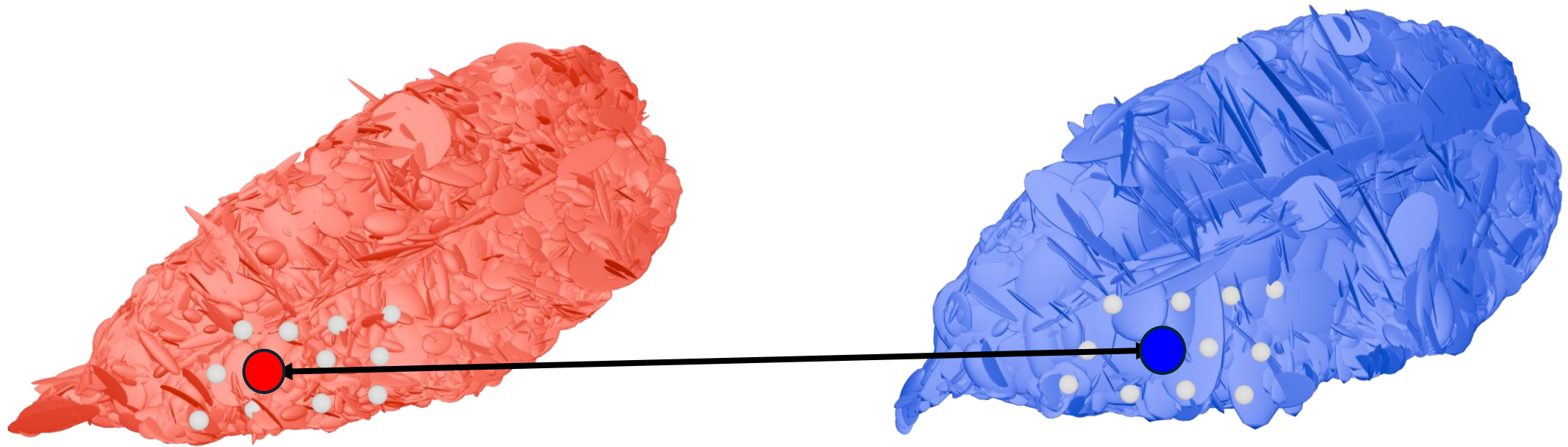
$$\sum_i w_i |T_i(p_i) - q_i|^2$$

Kernel Function Local Rigid Transformation

$$\frac{1}{|p_i - v|^{2\alpha}} \quad T(x) = Rx + t$$

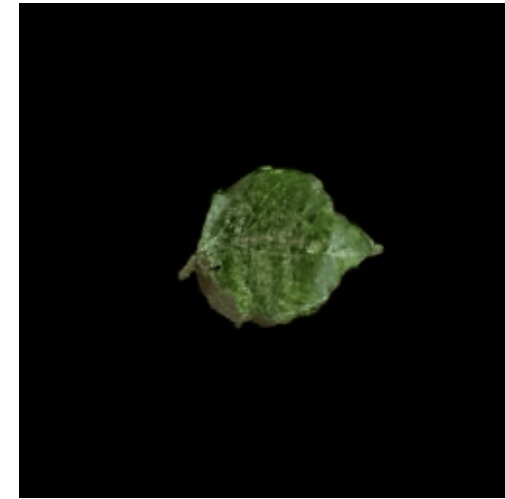
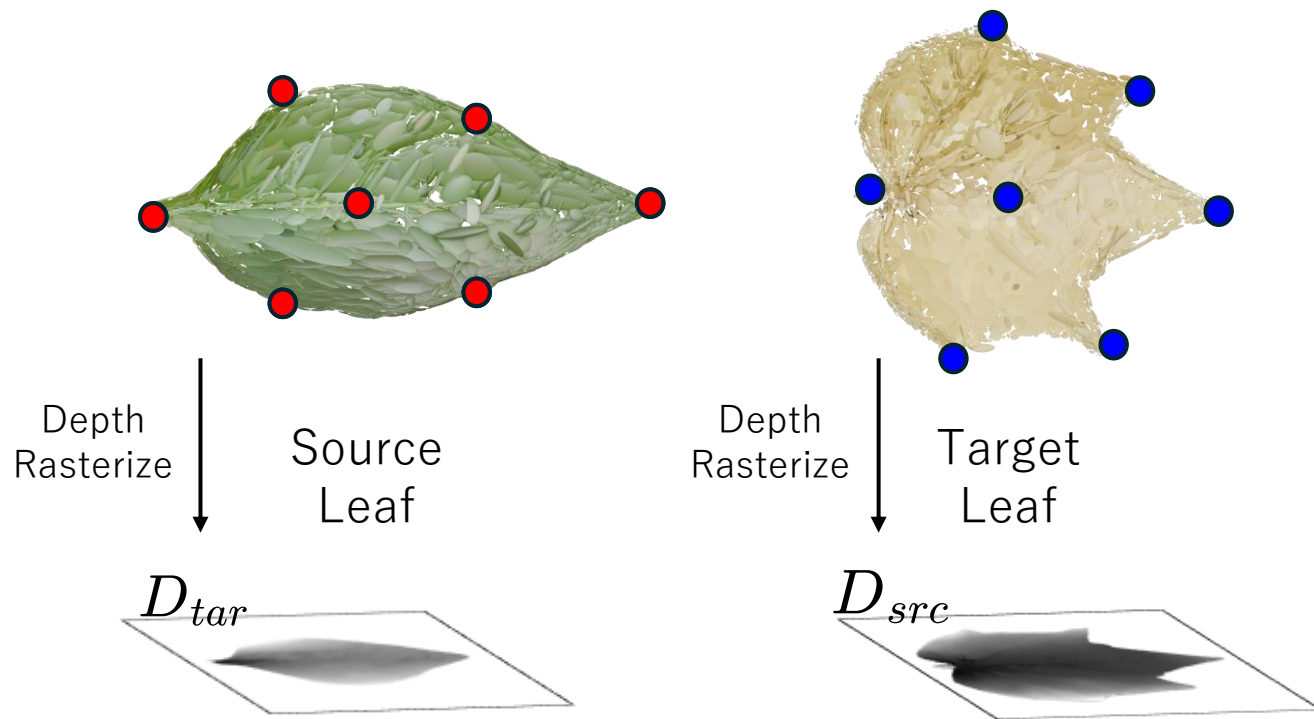
Initial Correspondence

- Align two leaves by PCA, build up initial correspondence.
- Farthest Point Sampling is applied to ensure uniform sampling.



Optimization

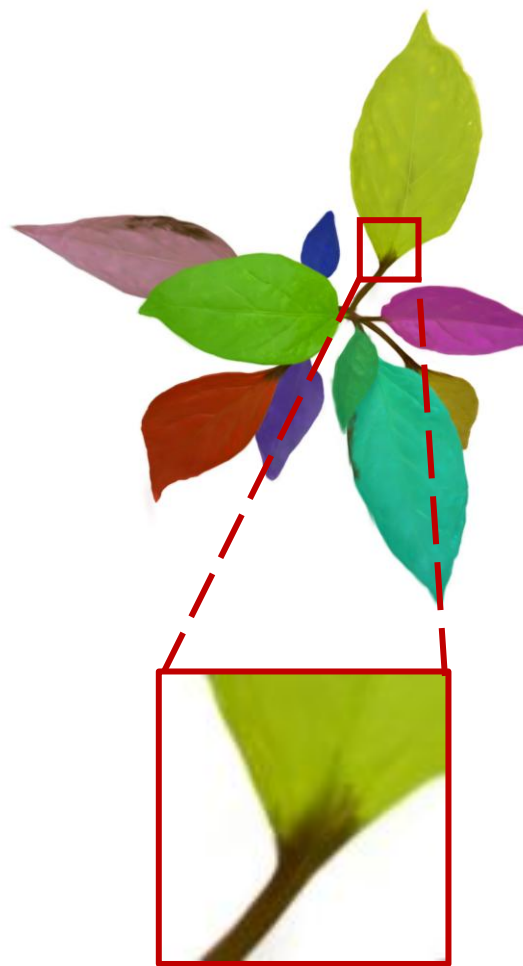
- Optimize correspondences with differentiable renderer.



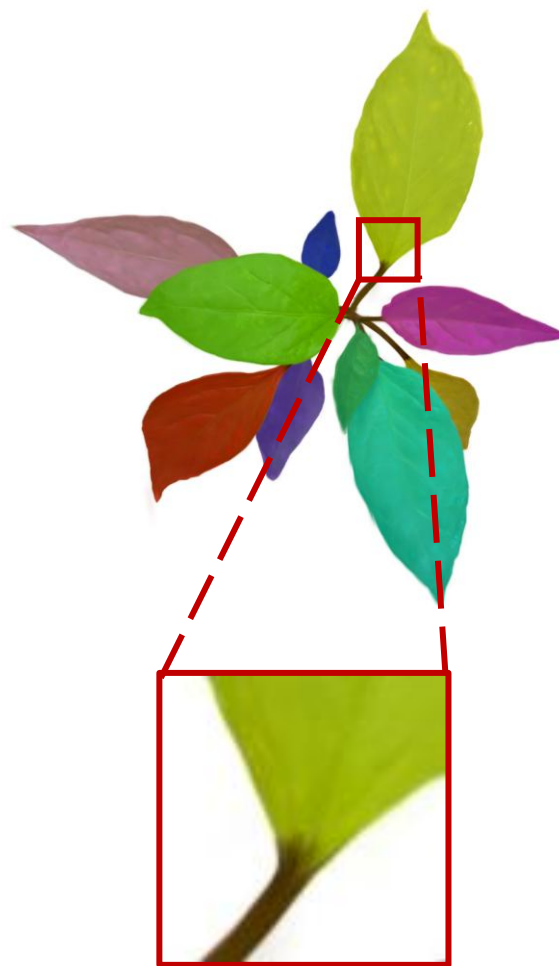
$$\mathcal{L}_{depth} = |D_{src} - D_{tar}|^2$$

$$\mathcal{L} = \mathcal{L}_{depth} + \lambda \mathcal{L}_{chamfer}$$

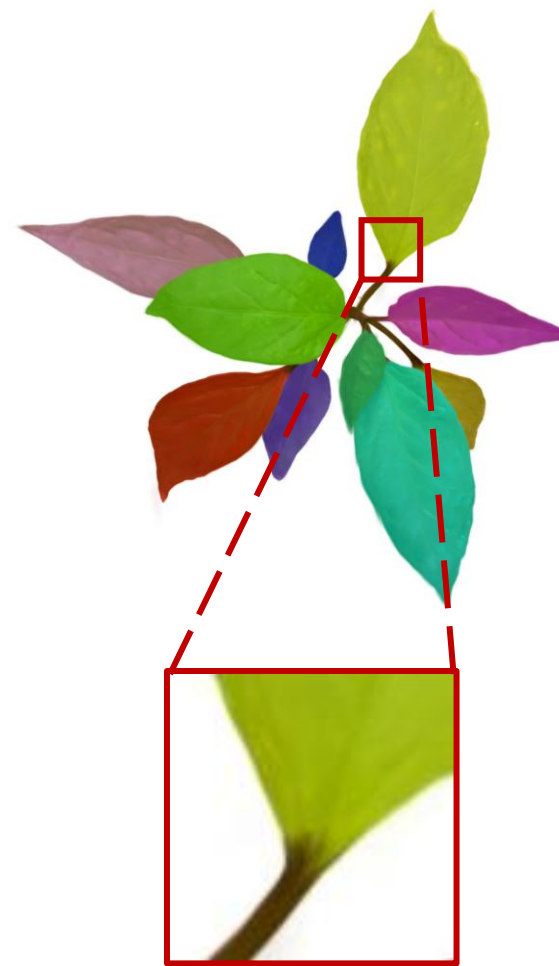
Segmentation comparison



Point Transformer

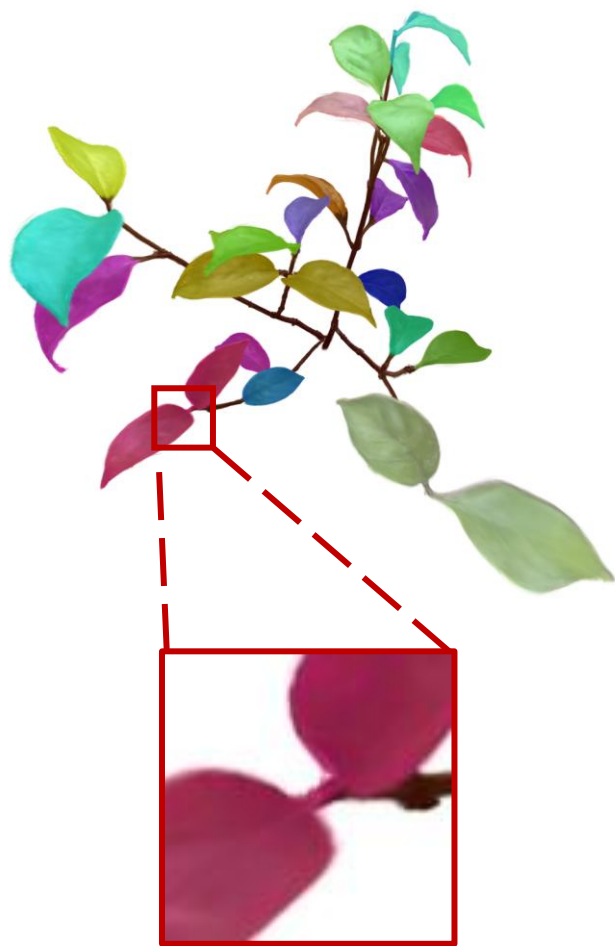


Ours

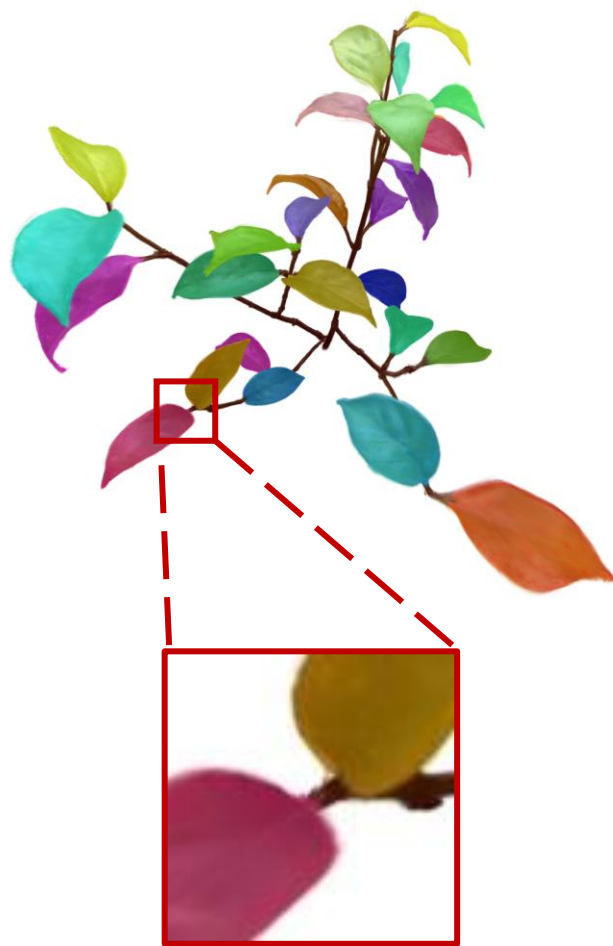


Ground truth

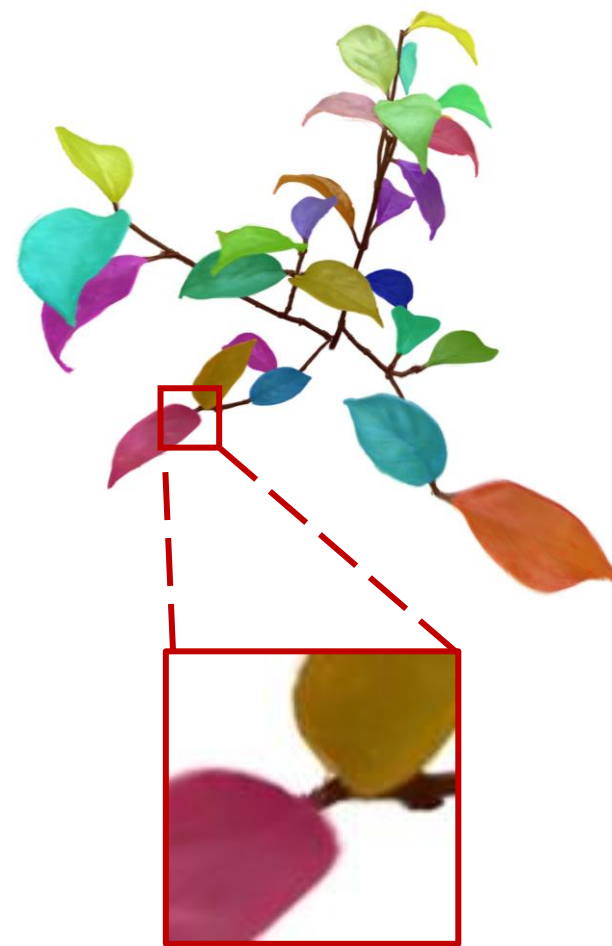
Segmentation comparison



Point Transformer



Ours



Ground truth

Mesh comparison

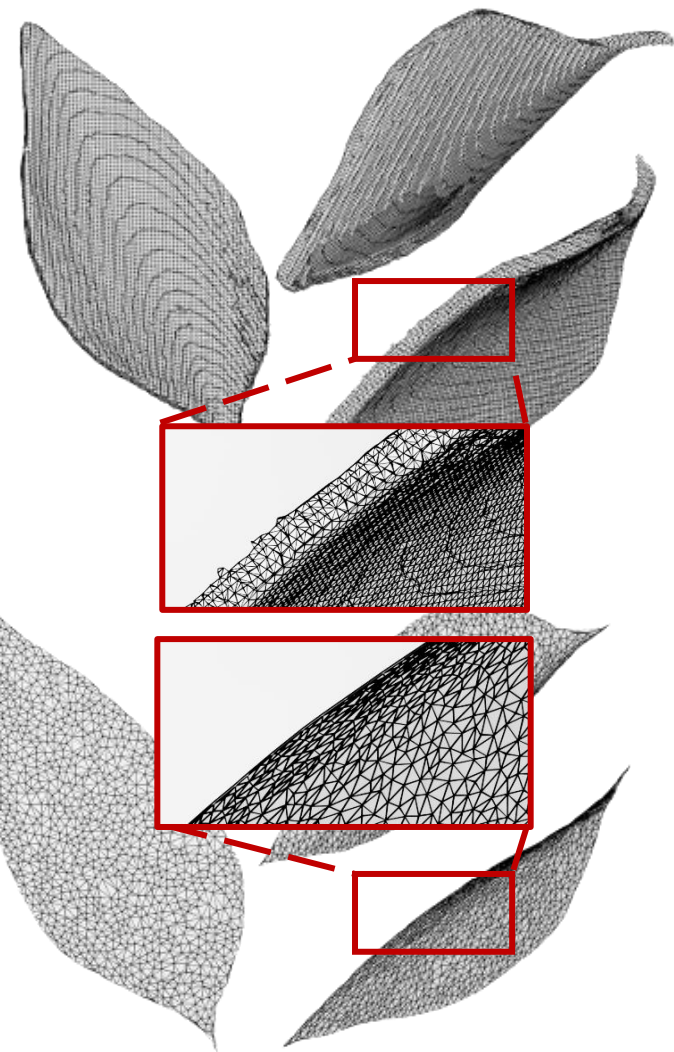
Input
Gaussian



GOF



Ours



Mesh comparison

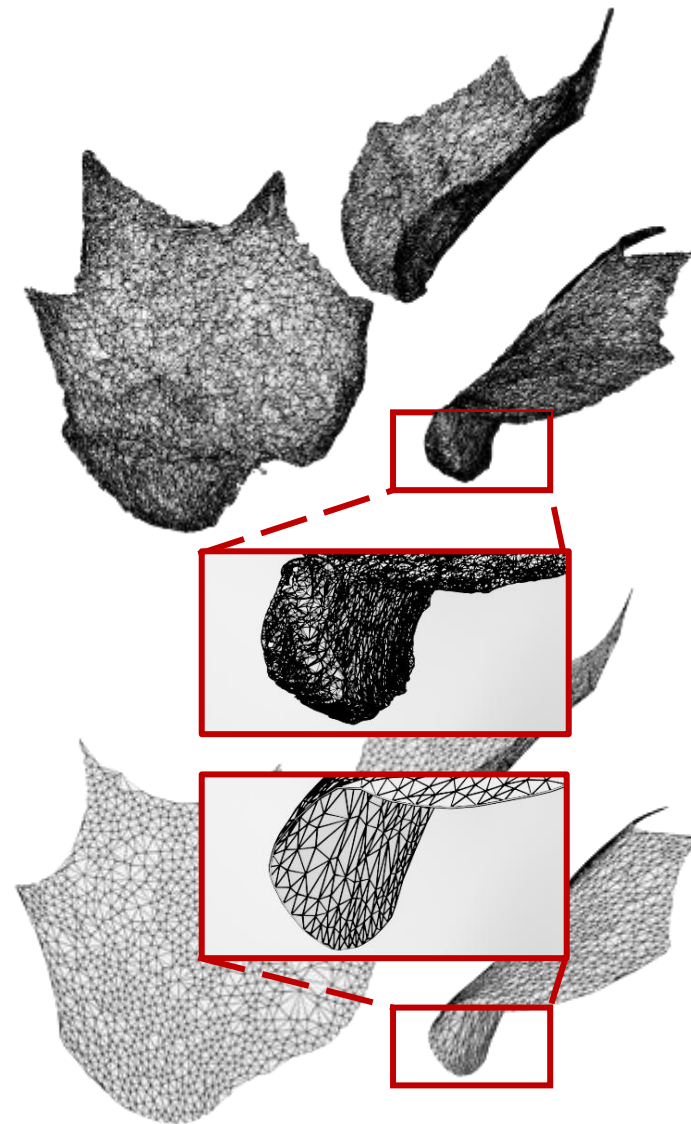
Input
Gaussian



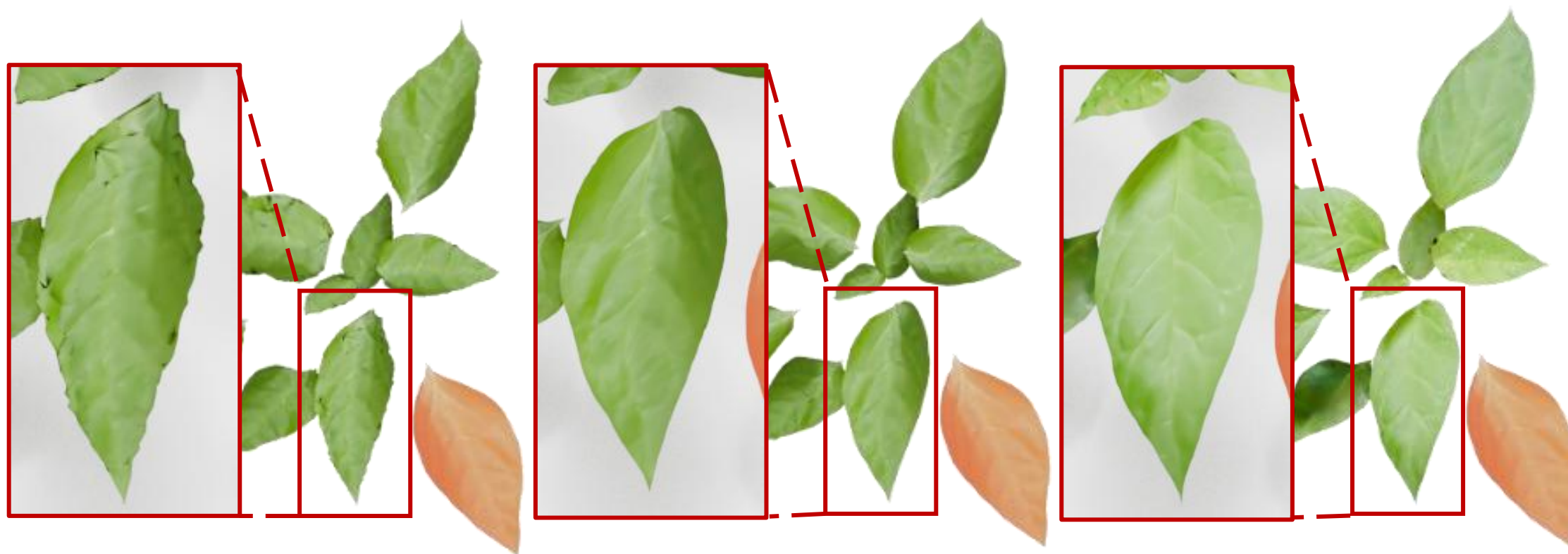
GOF



Ours



Deformation comparison



Non-Rigid ICP

Ours

Ground truth

Deformation comparison

PCA



NR-ICP



BCPD



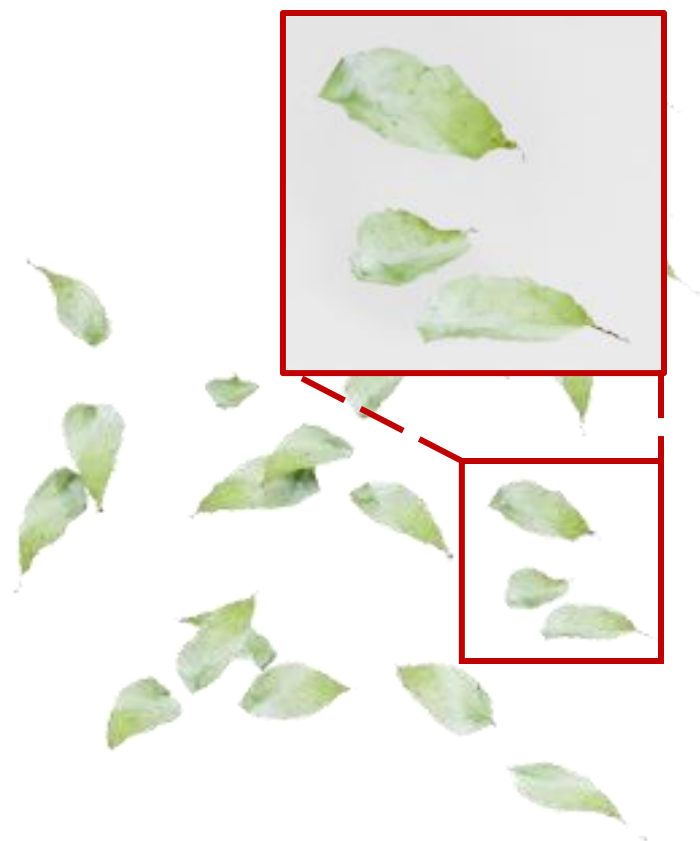
Ours



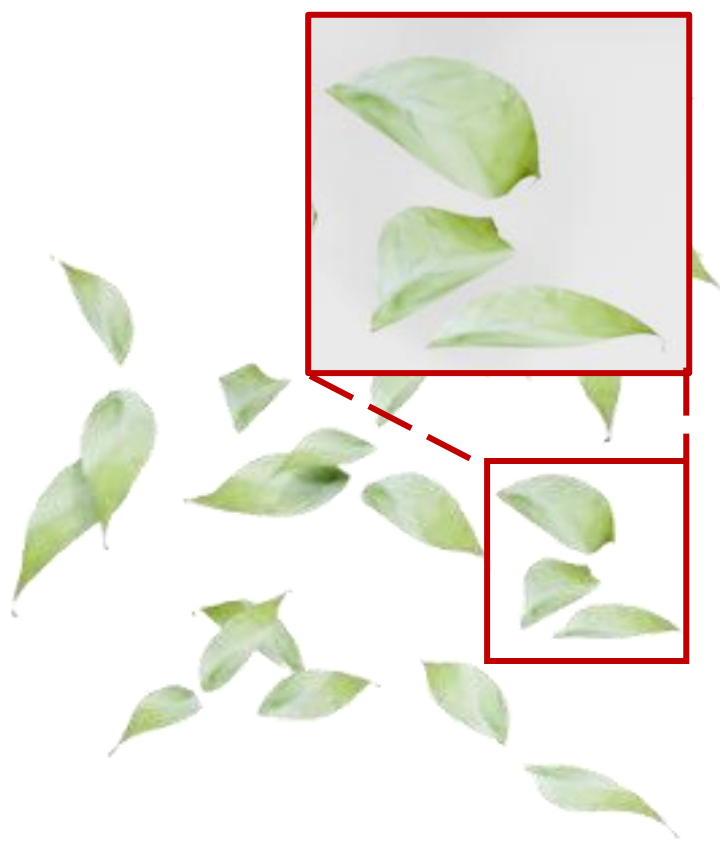
GT



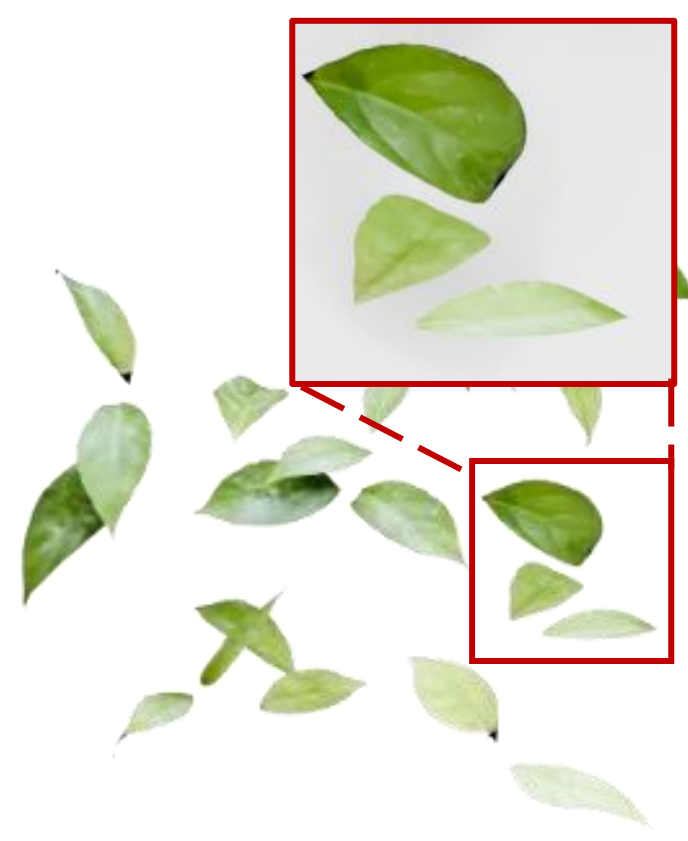
Deformation comparison



Non-Rigid ICP



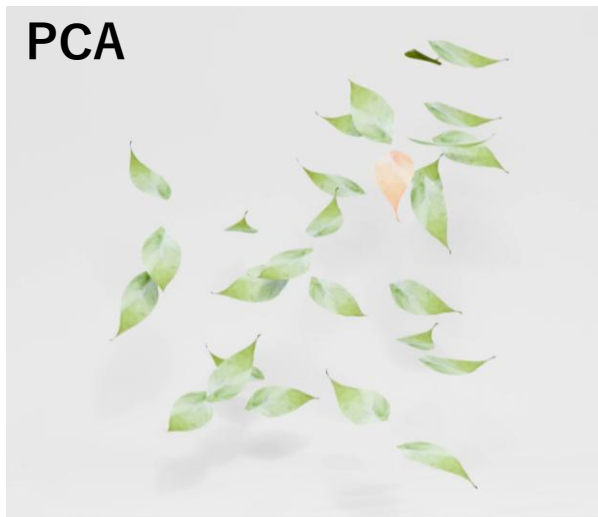
Ours



Ground truth

Deformation comparison

PCA



NR-ICP



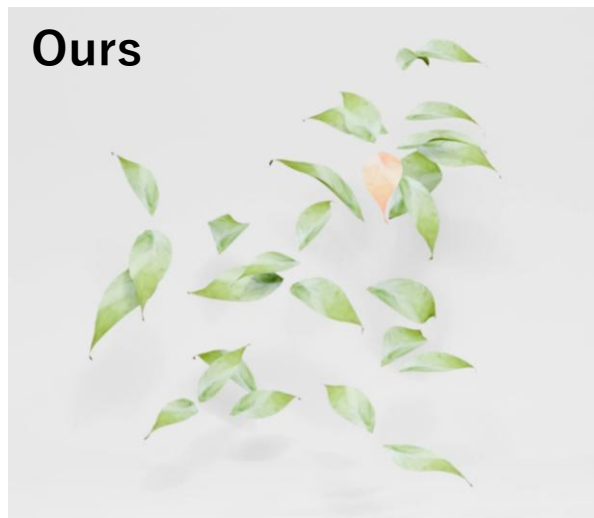
GT



BCPD



Ours



Results



(Gaussian Rendering)



(Ellipsoid Rendering)



GOF

Size: 26.2MB
#Primitive: 623k



Ours

Size: 0.75MB
#Primitive: 2k

Input 3DGS

Size: 11.9MB
#Primitive: 47k

Results



(Gaussian Rendering)



(Ellipsoid Rendering)



Input 3DGS

Size: 30.16MB

#Primitive: 122k

GOF

Size: 78.3MB

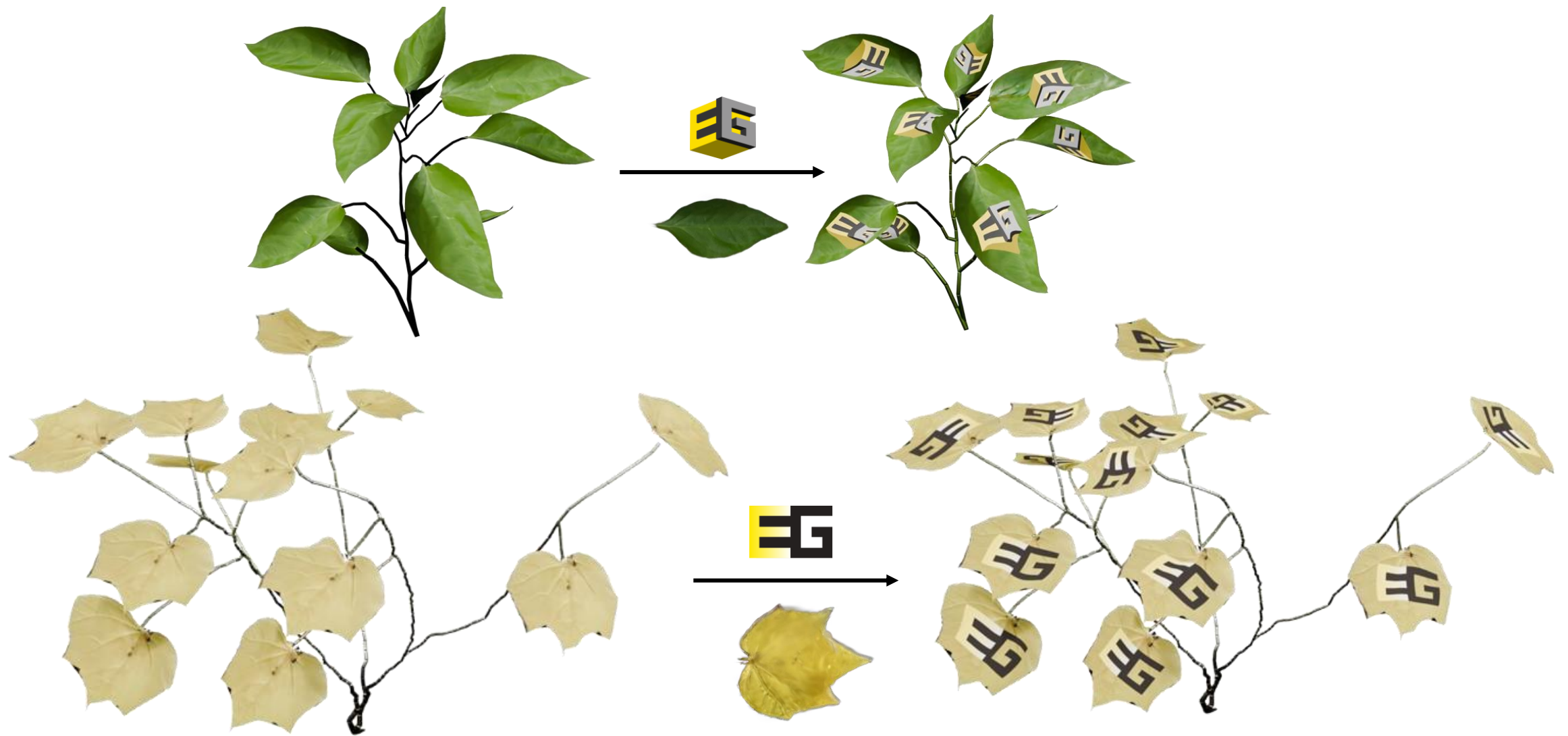
#Primitive: 1856k

Ours

Size: 1.86MB

#Primitive: 2k

Texture In-painting



Texture In-painting

